```
****************************************************
*                                                  *
*           STAAD.Pro V8i SELECTseries5            *
*           Version  20.07.10.65                   *
*           Proprietary Program of                 *
*           Bentley Systems, Inc.                  *
*           Date=    SEP  6, 2018                  *
*           Time=      8:43:19                     *
*                                                  *
*      USER ID:                                    *
****************************************************
```

```
   1. STAAD PLANE
INPUT FILE: 3f3k roof.STD
   2. START JOB INFORMATION
   3. ENGINEER DATE 16-NOV-17
   4. END JOB INFORMATION
   5. INPUT WIDTH 79
   6. UNIT METER KN
   7. JOINT COORDINATES
   8. 32 0 6.195 0; 33 0.606667 6.195 0; 34 1.21333 6.195 0; 35 1.82 6.195 0
   9. 36 2.42667 6.195 0; 37 3.03333 6.195 0; 38 3.64 6.195 0; 39 4.24667 6.195 0
  10. 40 4.85333 6.195 0; 41 5.46 6.195 0; 42 0 6.795 0; 43 0.303333 6.795 0
  11. 45 0.91 6.795 0; 47 1.51667 6.795 0; 49 2.12333 6.795 0; 51 2.73 6.795 0
  12. 53 3.33667 6.795 0; 55 3.94333 6.795 0; 57 4.55 6.795 0; 59 5.15667 6.795 0
  13. 60 5.46 6.795 0; 61 0 9.735 0; 62 0.91 9.735 0; 63 1.82 9.735 0
  14. 64 2.73 9.735 0; 65 3.64 9.735 0; 66 4.55 9.735 0; 67 5.46 9.735 0
  15. 68 0.91 9.945 0; 69 1.82 10.155 0; 70 2.73 10.365 0; 71 3.64 10.155 0
  16. 72 4.55 9.945 0; 74 5.46 9.385 0; 75 7.26 9.385 0; 76 6.66 6.795 0
  17. 78 6.375 6.795 0; 80 6.36 9.385 0; 82 6.36 9.56 0
  18. MEMBER INCIDENCES
  19. 3 32 42; 4 41 60; 5 61 42; 6 67 74; 11 74 60; 12 60 78; 14 78 76; 16 61 62
  20. 17 62 63; 18 63 64; 19 64 65; 20 65 66; 21 66 67; 22 61 68; 23 68 69; 24 69 70
  21. 25 70 71; 26 71 72; 27 72 67; 28 41 78; 30 62 68; 31 63 69; 32 64 70; 33 65 71
  22. 34 66 72; 35 63 68; 36 64 69; 37 64 71; 38 65 72; 39 67 82; 40 74 80; 41 80 75
  23. 42 82 75; 43 82 80; 44 74 82; 72 32 33; 73 33 34; 74 34 35; 75 35 36; 76 36 37
  24. 77 37 38; 78 38 39; 79 39 40; 80 40 41; 81 42 43; 90 43 45; 91 45 47; 92 47 49
  25. 93 49 51; 94 51 53; 95 53 55; 96 55 57; 97 57 59; 98 59 60; 117 32 43
  26. 118 43 33; 119 45 33; 120 45 34; 121 47 34; 122 47 35; 123 49 35; 124 49 36
  27. 125 51 36; 126 51 37; 127 53 37; 128 53 38; 129 55 38; 130 55 39; 131 57 39
  28. 132 57 40; 133 59 40; 134 59 41
  29. START USER TABLE
  30. TABLE 1
  31. UNIT MMS KN
  32. GENERAL
  33. 2C75X45
  34. 1098 75 6 90 3 938192 638334 100000 26218 14185 450 270 26218 14185 100000 75
  35. TABLE 2
  36. UNIT MMS KN
  37. GENERAL
  38. C75X45
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  39. 518 45 3 75 3 138000 449000 100000 4940 12000 270 315 4940 12000 100000 40
  40. TABLE 3
  41. UNIT MMS KN
  42. TUBE
  43. SHS38X3-CF
  44. 397 38 38 3 78500 78500 133000 228 152
  45. SHS50X3
  46. 564 50 50 3 208492 208492 311469 300 200
  47. SHS75X75X3
  48. 864 75 75 3 747792 747792 1.11974E+006 450 450
  49. TABLE 4
  50. UNIT MMS KN
  51. GENERAL
  52. C100X50
  53. 653 50 3 100 3 229000 990000 1E+006 7270 19800 300 343 11600 23600 1E+006 48
  54. TABLE 5
  55. UNIT MMS KN
  56. ANGLE
  57. UA38X38X3
  58. 38 38 3 7.51315 76 76
  59. TABLE 6
  60. UNIT MMS KN
  61. GENERAL
  62. C75NSHS75X3
  63. 1673 150 6 80 3 3.82575E+006 3.79732E+006 100000 51941 94933 690 848 -
  64. 51941 94933 100000 150
  65. END
  66. UNIT METER KN
  67. DEFINE MATERIAL START
  68. ISOTROPIC STEEL
  69. E 2.05E+008
  70. POISSON 0.3
  71. DENSITY 76.8195
  72. ALPHA 1.2E-005
  73. DAMP 0.03
  74. END DEFINE MATERIAL
  75. MEMBER PROPERTY BRITISH
  76. 16 TO 28 32 UPTABLE 2 C75X45
  77. MEMBER PROPERTY BRITISH
  78. 30 31 33 TO 44 UPTABLE 3 SHS38X3-CF
  79. MEMBER PROPERTY BRITISH
  80. 72 TO 81 90 TO 98 UPTABLE 4 C100X50
  81. MEMBER PROPERTY BRITISH
  82. 117 TO 134 UPTABLE 5 UA38X38X3
  83. MEMBER PROPERTY BRITISH
  84. 12 14 UPTABLE 3 SHS75X75X3
  85. MEMBER PROPERTY BRITISH
  86. 3 TO 6 11 UPTABLE 6 C75NSHS75X3
  87. CONSTANTS
  88. MATERIAL STEEL ALL
  89. SUPPORTS
  90. 41 PINNED
  91. 32 FIXED BUT FX FZ MX MY MZ
  92. MEMBER RELEASE
  93. 16 22 39 40 72 81 START MX MY MZ
  94. 21 27 80 98 END MX MY MZ
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 95. MEMBER TRUSS
 96. 30 TO 38 43 44 117 TO 134
 97. LOAD 1 LOADTYPE DEAD  TITLE DL
 98. SELFWEIGHT Y -1
 99. MEMBER LOAD
100. 12 14 81 90 TO 98 UNI GY -0.55
101. 22 TO 27 UNI GY -1.27
102. 39 42 UNI GY -0.64
103. LOAD 2 LOADTYPE LIVE REDUCIBLE TITLE LL
104. MEMBER LOAD
105. 12 14 81 90 TO 98 UNI GY -5.46
106. 22 TO 27 UNI GY -1.82
107. 39 42 UNI GY -0.91
108. LOAD 3 LOADTYPE WIND  TITLE WL
109. MEMBER LOAD
110. 16 TO 21 UNI GY 1.18
111. LOAD 4 LOADTYPE NONE  TITLE WLII
112. MEMBER LOAD
113. 3 5 UNI GX 0.6
114. LOAD COMB 5 1.35DL+1.5LL
115. 1 1.35 2 1.5
116. LOAD COMB 6 1.35DL+1.5LL+0.75WL
117. 1 1.35 2 1.5 3 0.75
118. LOAD COMB 7 1.35DL+1.05LL+1.35WL
119. 1 1.35 2 1.05 3 1.35
120. LOAD COMB 8 1.35DL+1.5LL+0.75WLII
121. 1 1.35 2 1.5 3 0.75
122. LOAD COMB 9 1.35DL+1.05LL+1.35WLII
123. 1 1.35 2 1.05 4 1.35
124. LOAD COMB 10 1.0DL+1.0LL
125. 1 1.0 2 1.0
126. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

   NUMBER OF JOINTS          39  NUMBER OF MEMBERS      72
   NUMBER OF PLATES           0  NUMBER OF SOLIDS        0
   NUMBER OF SURFACES         0  NUMBER OF SUPPORTS      2

         SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=    27/      5/     18 DOF
 TOTAL      PRIMARY LOAD CASES =      4, TOTAL DEGREES OF FREEDOM =      114
 TOTAL LOAD COMBINATION  CASES =      6  SO FAR.
   SIZE OF STIFFNESS MATRIX =         3 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.1/-743996.7 MB
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127. PARAMETER 1
128. CODE EN 1993-1-1:2005
129. SGR 1 ALL
130. CHECK CODE ALL

                    STAAD.PRO CODE CHECKING - EN 1993-1-1:2005
                    *******************************************
                    NATIONAL ANNEX - NOT USED

PROGRAM CODE REVISION V1.11 BS_EC3_2005/1
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